# How BARCS Finds Important Genes

Answer sheet: Chapter 2 exercises

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BARCS version 0.2.2 · Reading level: grades 7 to 9

Each answer shows the calculation first. "Why this matters" explains the idea behind the question. Values: $\mu = 0.00015$ (control), $\mu = 0.000075$ (drug), $\rho = 0.000001$, $\beta_1 = -0.69$.

## Part A: Fitting

**1.** (a) $0.00015 \times 0.99985 = 0.000149978$. Slope: $1 \div 0.000149978 = 6{,}668$.

(b) Step: $(0.00018 - 0.00015) \times 6{,}668 = 0.00003 \times 6{,}668 = 0.200$. Working value: $v = -8.8047 + 0.200 = -8.6047$.

(c) $0.00018 \div 0.99982 = 0.000180032$. $\ln(0.000180032) = -8.6224$. The difference is only 0.018.

(d) With $K = 0$, the real log-odds is $\ln(0)$, which does not exist. The working value still exists: $v = -8.8047 + (0 - 0.00015) \times 6{,}668 = -8.8047 - 1.000 = -9.805$. The straight line also makes the variance of $v$ easy to calculate, which gives the weight.

**Why this matters.** The working value lets BARCS use simple weighted averages, even for guides with zero reads.

**2.** (a) Library with 1 million reads: top $= 1{,}000{,}000 \times 0.00015 \times 0.99985 = 149.98$; bottom $= 1 + 999{,}999 \times 0.000001 = 2.0$; $w = 149.98 \div 2.0 = 75.0$.
Library with 2 million reads: top $= 299.96$; bottom $= 1 + 1{,}999{,}999 \times 0.000001 = 3.0$; $w = 299.96 \div 3.0 = 100.0$.

(b) No. Twice the reads gives only $100 \div 75 = 1.33$ times the weight. The extra reads shrink the counting noise, but the biological noise of the flask stays. The bottom of the weight grows from 2 to 3.

(c) With $\rho = 0$, the bottom is 1. The weights are 150 and 300, exactly double.

**Why this matters.** The weight says how much a library can tell you. Depth helps only until biological noise dominates.

**3.** (a) $(75 \times (-8.60) + 100 \times (-8.90)) \div (75 + 100) = (-645 - 890) \div 175 = -1{,}535 \div 175 = -8.771$.

(b) $\sqrt{1 \div 175} = \sqrt{0.005714} = 0.0756$.

(c) The weight is $1 \div$ variance. The library with weight 100 has less noise, so it should count more. "Weight $= 1 \div$ variance" gives the average with the smallest possible variance.

**Why this matters.** A weighted average trusts the precise libraries more. This is the core of the BARCS fit.

# Part B: Is the effect real?

**4.** (a) $1 \div 175 = 0.005714$; $1 \div 70 = 0.014286$; sum $= 0.020000$; SE $= \sqrt{0.020000} = 0.1414$.

(b) $t = -0.69 \div 0.1414 = -4.88$. df $= 4 - 2 = 2$.

(c) $|t| = 4.88$ is larger than 4.30. So $p < 0.05$ (a computer gives $p = 0.040$). It is significant, but only just.

(d) The drug group: 0.014286 of the 0.020000 is from the drug group (71%). Its total weight is small because its libraries are smaller and its proportion is lower. Add drug replicates. That adds weight where it is missing, and it adds degrees of freedom.

**Why this matters.** The SE shows where the uncertainty comes from. That tells you how to design the next experiment.

**5.** (a) Guide P: $5.1 \div 2 = 2.55$. Guide Q: $1.4 \div 2 = 0.70$.

(b) Guide P has a sum above 2, so it has more noise than counting noise: BARCS uses $\rho > 0$. Guide Q has a sum of 2 or less: BARCS uses $\rho = 0$.

(c) The predictions use two $\beta$ values that were fitted to the same four libraries. The fit pulls the predictions toward the data, and each fitted $\beta$ takes away 1 from the sum on average. So the expected sum is $4 - 2 = 2$, the degrees of freedom.

**Why this matters.** BARCS measures $\rho$ by making the noise model agree with the data on average.

**6.** (a) $(2 \times 2.55 + 10 \times 1.5) \div (2 + 10) = (5.1 + 15) \div 12 = 20.1 \div 12 = 1.675$.

(b) $\sqrt{1.675 \div 2.55} = \sqrt{0.657} = 0.81$. The SE becomes 19% smaller. New degrees of freedom: $2 + 10 = 12$. Now $|t|$ needs to be larger than only 2.18 (not 4.30) for $p < 0.05$.

(c) The guides of one screen went through the same cells, the same library preparation and the same sequencer. So their noise is similar. One guide with 2 degrees of freedom gives a very unsure noise estimate, and the other guides make it steadier. It can be a bad idea when a guide really is much noisier than the rest, for example when a few cell clones take over a flask. Then borrowing makes that guide look too precise. The check with control guides (Exercise 7) catches this problem.

**Why this matters.** Moderation turns thousands of guides into extra information about noise. That is how a small experiment can still find real effects.

## Part C: From guides to a list of genes

**7.** (a) $0.05 \times 300 = 15$. The real number is 45, three times more. The noise model is too small: the libraries are noisier than the model thinks. One common cause is replicates that are not independent, for example flasks split from one population of cells.

(b) $2.6 \div 1.96 = 1.33$.

(c) $3.0 \div 1.33 = 2.26$. It is still beyond 1.96, so the gene keeps $p < 0.05$ (with many degrees of freedom).

**Why this matters.** Control guides have no true effect. They are a free test of the noise model inside your own experiment.

**8.** (a) $(2.2 + 1.8 - 0.3 + 2.5) \div \sqrt{4} = 6.2 \div 2 = 3.1$ ($p = 0.002$).

(b) Bottom: $\sqrt{4 + 4 \times 3 \times 0.2} = \sqrt{4 + 2.4} = \sqrt{6.4} = 2.53$. $Z = 6.2 \div 2.53 = 2.45$ ($p = 0.014$).

(c) A negative $z$ means that this guide moved the other way. It is evidence against a consistent effect, so it lowers the sum. With shared noise, the guides agree partly because they share noise, not because of the gene. Each of the $4 \times 3 = 12$ pairs adds $r$ to the variance of the sum (squares and pairs, Chapter 1). A bigger variance means a bigger bottom and a smaller $Z$.

**Why this matters.** A gene is believable when its independent guides agree. Shared noise makes agreement less convincing.

**9.** (a) $10{,}000 \times 0.0005 = 5$ genes.

(b) About $5 \div 40 = 0.125$. About one gene in eight on this list is probably false.

(c) Use a smaller limit. Fewer genes pass by chance, and the list usually becomes cleaner, but also shorter. Benjamini–Hochberg finds the limit for you.

**Why this matters.** With thousands of tests, a few small p-values happen by chance. The FDR tells you how many of your hits to expect to be false.

**10.** The extra factor at 20 million reads is $1 + 19{,}999{,}999 \times 0.000001 \approx 21$. So the biological noise is about 20 times the counting noise. The standard deviation of the proportion is 12.5 per million, almost the same as the limit of 12.2 per million (Chapter 1, Exercise 8), and not much better than 17.3 per million at 1 million reads. The deep reading buys almost nothing. Also, 2 replicates per group give 4 libraries and only $4 - 2 = 2$ degrees of freedom, so $|t|$ must be larger than 4.30. A better plan: more replicates per group (for example 4) with fewer reads each.

**Why this matters.** Reads and replicates are not the same thing. Degrees of freedom and biological noise come from replicates only.

**11.** (a) $(2.0 + 2.6 + 2.2) \div 3 = 6.8 \div 3 = 2.27$.

(b) $p = 2 \times (1 + 9) \div (1 + 1{,}999) = 2 \times 10 \div 2{,}000 = 0.01$.

(c) $\sqrt{1 + (3 - 1) \times 0.1} = \sqrt{1.2} = 1.10$. The guides of a real gene share some noise, so their average wobbles more than the average of unrelated control guides. Without the widening, the pseudo-genes would look calmer than a real gene with no effect, and real genes would look too significant.

(d) $2 \div (150 + 1) = 2 \div 151 = 0.013$. No one-guide gene can get a smaller p-value, and the FDR step needs many genes at that floor. So "no hits" here means "150 controls are too few to confirm the hits", not "nothing changed". Add more control guides to the next library, or look at genes with several guides.

**Why this matters.** The empirical null trusts the controls of your own experiment. Its strength (no model needed) is also its limit (it can only be as fine as the number of controls).